

CSC 103 Final Exam Cheat Sheet

Declaring a variable

```
int x;
```

Initializing a variable

```
x = 10;
```

Using a for loop

This example is a for loop that will run its code 50 times:

```
for (int i=0; i<50; i=i+1){  
    *Code to run for each iteration*  
}
```

Using a while loop

This is a while loop that will run 50 times:

```
int i = 0;  
while (i < 50) {  
    i = i + 1;  
}
```

mousePressed function

```
void mousePressed(){  
    *code to run whenever the mouse is clicked*  
}
```

keyPressed function

```
void keyPressed(){  
    *code to run whenever a key is pressed*  
}
```

If statements

Example of using booleans:

```
if (GraceIsAwesome == true){  
    println("Hooray for Grace!");  
} else if (GraceHasCookies == true){  
    println("Grace, give me cookies!");  
} else {  
    println("Grace is not awesome and does not  
    have cookies 😞");  
}
```

Example of using Boolean expression:

```
if (x < 0){  
    println("x is small");  
}
```

Defining a function

```
*function type* *function name*(*inputs*){  
    *code to run in function*  
}
```

Example of a function that draw a red rectangle at the specified location with a specified width and height:

```
void redRectangle(int rX, int rY, int rW, int rH){  
    fill(255,0,0);  
    rect(rX, rY, rW, rH)  
}
```

Boolean operators

&& means “AND”

|| means “OR”

! means “NOT”

> means “greater than”

>= means “greater than or equal to”

< means “less than”

<= means “less than or equal to”

== means “equal to”

!= means “not equal to”

Other helpful functions and commands

- random(num1, num2): generates a random number between num1 and num2:

- Ex: This line of code:
“a = random(0, 100);” would make “a” equal to a random number between 0 and 100.

- rect(x, y, w, h): draws a rectangle such that the rectangle’s top left corner is at the point (x,y). The width of the rectangle is controlled by w, and its height is controlled by h.

- triangle(x1, y1, x2, y2, x3, y3): draws a triangle between the points (x1,y1), (x2, y2), and (x3, y3).

- size(w,h): makes the window have a width of w and a height of h.

-background(c): makes the background have the color c

-circle(x,y,d): draw a circle at point x,y with a diameter of d.

-noFill(): makes it so that anything that is drawn after this command does not have a fill.

-fill(R, G, B): makes it so that the fill of any shape drawn after this command will have a color specified by the given RGB values.

-stroke(R, G, B): makes it so that the stroke of any shape drawn after this command will have a color specified by the given RGB values.

- strokeWeight(num): makes the thickness of the stroke be equal to num.